

Assignment No- 04

Subject- Operating System

1	Write a short note on memory management.
2	Explain contiguous memory management technique.
3	Write a short note on Belady's anomaly.
4	Differentiate paging and segmentation.
5	Explain the placement algorithm with suitable example.
6	Explain FIFO with suitable example.
7	Explain LRU with suitable example.
8	Explain Optimal page replacement algorithm with suitable example.
9	Write a short note on virtual memory.
10	Explain thrashing with suitable example. And also explain method to avoid thrashing.
11	Explain non-contiguous memory management technique.
12	Explain internal fragmentation, external fragmentation & compaction with suitable example.
13	Why are segmentation and paging sometimes combined into one scheme?
14	What is copy-on write feature and under what circumstances it is beneficial?
15	Explain the purpose of paging the page table?
16	What is the main function of the memory-management unit?
17	Write short on: a. multiprogramming with fixed & variable partition.
18	What is virtual memory? How it is implemented
19	What is demand paging? Explain it with address translation mechanism used. What are its
20	Explain hierarchical page table and inverted page table.
21	Explain static partitioned allocation with partition sizes 300,150, 100, 200, 20. Assuming first fit method indicate the memory status after memory request for sizes 80, 180, 280, 380, 30
22	Write short note on: Relocation problem for multiprogramming with fixed partitions.
23	Consider the following reference string 7,0,1,2,0,3,0,4,2,3,0,3,2,1,2,0,1,7,0,1. Assume there are three frames. Apply LRU replacement algorithm to the reference string above and find out how many page faults are produced. Illustrate the LRU page replacement algorithm in detail and also two feasible implementation of the LRU algorithm
24	Explain about Swapping.
25	Discuss the procedure for page fault in demand paging.
26	Explain Page Replacement Strategies.

2 7	Explain the difference between logical and physical addresses?
2 8	What do you mean by thrashing?
2 9	Distinguish between page and segment.
3 0	How the problem of external fragmentation can be solved.
3 1	What you mean by compaction? In which situation is it applied.
3 2	What are the major problems to implement Demand Paging?
3 3	Define Secondary Memory.
3 4	Consider the following page reference string: 1, 2, 3, 4, 2, 1, 5, 6, 2, 1, 2, 3, 7, 6, 3, 2, 1, 2, 3, 6. Identify the no. of page faults would occur for the following replacement algorithms, assuming one, two, three, four, five, six, or seven frames? Remember all frames are initially empty, so your first unique pages will all cost one fault each. a. LRU replacement b. FIFO replacement c. Optimal replacement
3 5	What do you mean by Best Fit placement algorithm explain with the help of suitable example?
3 6	What do you mean by First Fit placement algorithm explain with the help of suitable example?
3 7	What do you mean by Worst Fit placement algorithm explain with the help of suitable example?
3 8	What do you mean by Next Fit placement algorithm explain with the help of suitable example?
3 9	Explain in briefly and compare, fixed and dynamic memory partitioning schemes.
4 0	Consider the following page reference string: 1,2,3,2,5,6,3,4,6,3,7,3,1,5,3,6,3,4,2,4,3,4,5,1. Indicate page faults and calculate total number of page faults and successful ratio for FIFO, optimal and LRU algorithms. Assume there are four frames and initially all the frames are empty.